

Homework 3

1. Prove or disprove the following statements.

(a) Suppose f is a bounded real function on $[a, b]$, and f^2 is Riemann integrable on $[a, b]$, then f is Riemann integrable on $[a, b]$

(b) Suppose f is a bounded real function on $[a, b]$, and f^3 is Riemann integrable on $[a, b]$, then f is Riemann integrable on $[a, b]$

2. Prove: If f is continuous on $[a, b] \times [c, d]$, then the function

$$F(y) = \int_a^b f(x, y) dx$$

is continuous on $[c, d]$.

3. Let

$$f(x, y) = \begin{cases} 2xy & \text{if } y \text{ is rational,} \\ y & \text{if } y \text{ is irrational} \end{cases}$$

and $\mathbf{R} = [0, 1] \times [0, 1]$. Calculate $\iint_{\mathbf{R}} f(x, y) d(x, y)$ and $\overline{\iint_{\mathbf{R}}} f(x, y) d(x, y)$, and show that f is not integrable on $\mathbf{R}$.

4. Prove: If $f : [a_0, b_0] \rightarrow \mathbb{R}^2$ is of class $\mathbf{C}^1$, then $f([a, b])$ has zero content whenever $a_0 < a < b < b_0$.

5. Prove: If $f : \mathbb{R}^k \rightarrow \mathbb{R}$ is Riemann integrable on S and on T , and $S \cap T$ is content zero, then f is Riemann integrable on $S \cup T$, and $\int_{S \cup T} f dV^k = \int_S f dV^k + \int_T f dV^k$.